Higher Pay or Smaller Classes? Retaining Teachers in Hard-to-Staff Schools

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Abstract

This paper studies the effects of higher pay and better working conditions on teacher retention in disadvantaged schools. I exploit two French reforms: a financial bonus introduced in 2015 and subsequently increased for the most disadvantaged schools, and a class size reduction progressively rolled out from 2017, capping first- and second-grade classes at approximately half their pre-reform size. Because some teachers were exposed to the bonus alone while others were also exposed to class size reduction, comparing retention across the two groups separately identifies the effect of each instrument. Using an event study design, I find that the bonus exhibits a threshold: a bonus below approximately 6% of annual salary has no detectable effect on retention, while a bonus reaching approximately 10% increases retention by 2 to 3 percentage points. Class size reduction generates further gains among teachers exposed to both instruments, which together produce the largest effect in the paper, up to 4 percentage points. These findings show that higher pay and better working conditions are both powerful levers for stabilizing the teaching workforce in disadvantaged schools, and that their combined effect exceeds what either can achieve alone.

Keywords: Class size, teacher bonus, education policy, priority education

JEL Classification: I21, I28, J45, J63

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1 Introduction

Attracting and retaining teachers in hard-to-staff schools serving disadvantaged populations remains a persistent challenge in most education systems, largely because of the demanding working conditions these schools present (Hanushek, Kain & Rivkin, 2004; Boyd et al., 2005). Evidence shows that higher pay can improve teacher retention in these schools (Clotfelter et al., 2008; Steele et al., 2010). However, less is known about whether better working conditions can achieve retention effects similar to those of higher pay, and whether combining the two produces larger gains than either instrument alone.

Two major reforms targeting hard-to-staff schools in France offer a rare opportunity to compare the relative effects of higher pay and better working conditions on teacher retention. The first raises teacher pay: in 2015, a substantial bonus was introduced for teachers working in disadvantaged schools, subsequently increased further in a specific subset of these schools in 2018, 2019 and 2021. The second improves working conditions: starting in 2017, a class size reduction policy was progressively rolled out in disadvantaged primary schools, capping first- and second-grade classes at a maximum of 12 students, approximately half the pre-reform average. The explicit objective of this reform was to reduce educational inequalities from the earliest age. Yet, by reducing class sizes, the policy may also have made teaching in disadvantaged schools less demanding, and potentially more attractive to remain in.

This paper exploits the staggered timing and differential intensity of these two reforms across pre-primary and primary grades to separately identify the retention effects of higher pay and class size reduction. My identification strategy relies on the organisation of the French primary school system around pre-primary grades, covering ages 3–6, and primary grades, covering ages 6–11. Because class size reductions targeted first- and second-grade classrooms, they applied to primary grades from 2017 onwards but left pre-primary grades largely unaffected until 2020. By contrast, the bonus applied equally to teachers in both pre-primary and primary grades. This offers a unique setting for a quasi-experimental design in which pre-primary grade teachers were exposed to higher pay only, while primary grade teachers were exposed to both higher pay and class size reduction, with the intensity of both instruments differing across specific types of disadvantaged schools. I use an event study design comparing the evolution of retention rates in disadvantaged schools with retention rates in regular public schools, estimated separately for pre-primary and primary grade teachers. Identification relies on a parallel trends assumption: absent the reforms, retention rates in disadvantaged and regular public schools would have evolved similarly over time, an assumption supported by pre-trend tests reported throughout the paper.

I find that the bonus alone increases retention, and that combining it with class size reduction generates the largest effects, reaching up to 4 percentage points. Comparing schools targeted by the reforms to regular public schools, teachers exposed only to the financial bonus show retention gains of approximately 2–3 percentage points, while teachers exposed to both the bonus and class size reduction show gains of 2 to 4 percentage points, growing over time as the policy was progressively expanded. The larger and more sustained effect among teachers exposed to both instruments is consistent with class size reduction amplifying the retention effect of the financial bonus. The financial bonus also exhibits a threshold: a bonus below approximately 6% of annual salary generates no significant retention gain, while a bonus reaching approximately 10% of annual salary does, even absent any class size reduction, confirming that a sufficiently large financial premium can independently stabilize the teaching workforce.

The estimated effects are likely conservative along two dimensions. First, because the administrative data do not record each teacher's specific grade assignment, class size reduction estimates are computed on all primary grade teachers. However, the policy specifically targeted first- and second-grade classrooms, covering only two of five grade levels in a typical primary school. For a subset of teachers whose teaching grade could be recovered through an auxiliary matching procedure, descriptive evidence suggests the true effect on directly treated teachers is larger than these estimates imply. Second, these effects are estimated on tenured teachers holding a permanent post, a stable segment of the teaching workforce whose observed mobility closely approximates their stated preferences and whose baseline retention rate already exceeded 85% before the reforms; effects on more mobile teachers, who were not the focus of this analysis, could be substantially larger.

Taken together, these findings speak to the broader question of how education systems can retain teachers in disadvantaged schools, and to which policy instruments, financial or otherwise, are best suited to do so.

1.1 Related Literature

This paper relates to three strands of the literature. The first concerns the effect of financial incentives on teacher retention in disadvantaged schools. Existing evidence documents positive effects of geographically targeted bonuses: Clotfelter et al. (8) find that a bonus in North Carolina reduced teacher turnover in high-poverty schools, and Falch (12) finds that a 10% wage increase lowers voluntary exits in hard-to-staff Norwegian schools. These studies, however, evaluate a single bonus level and cannot speak to whether the magnitude of the incentive matters for its effectiveness. This paper contributes to this literature by identifying a threshold in the size of the bonus: a small, stagnant bonus generates no detectable retention response over several years, while a substantially larger bonus does, consistent with the difficulty of using small incentives to offset otherwise unattractive working conditions in disadvantaged schools (10).

Second, this paper relates to a smaller literature on the effect of class size reduction on teacher retention specifically. Class size reduction has been extensively studied for its effects on student achievement, both in randomized (19) and quasi-experimental settings (27; 32; 16; 20), but its effect on teachers themselves, and on their retention decisions in particular, has received comparatively little attention. The broader literature on teacher working conditions documents that working conditions more generally are associated with teacher turnover (21; 17), but without isolating class size reduction as a specific, identifiable policy lever. This paper provides, to the best of my knowledge, one of the first pieces of causal evidence on the effect of class size reduction on teacher retention. I further show that class size reduction reinforces the retention effect of a bonus even when that bonus alone is insufficient to generate substantial retention gains, indicating that better working conditions and higher pay can complement one another even at modest levels of either instrument.

Third, this paper contributes to the broader literature on teacher labor markets and mobility in disadvantaged schools (28; 1; 5). Documenting how teachers respond to simultaneous improvements in the financial and non-financial dimensions of their working conditions, I show that the two instruments reinforce one another where both were introduced, with the combination generating the largest retention gains observed in this paper. This speaks to the broader question of how compensatory education policies can be designed to retain teachers in the schools that need them most.

The rest of the paper is organized as follows: Section 2 describes the institutional context, Section 3 presents the data, Section 4 outlines the empirical strategy, Section 5 reports the results, and Section 6 concludes.

2 Institutional Setting

2.1 Priority Education

Priority education networks were created in 1981 to reduce educational inequalities by allocating additional resources to schools serving areas with high concentrations of disadvantaged students. The policy's stated objective has remained consistent across its various reforms: improving academic outcomes for the most disadvantaged students by compensating for unequal socioeconomic conditions across catchment areas. Initially organized around *zones d'éducation prioritaire* (ZEP), the program underwent several reorganizations over subsequent decades, evolving into networks structured around lower secondary schools (*réseaux d'éducation prioritaire*) and their feeder primary schools.

The classification relevant to this study dates from the 2014–2015 reform, which restructured the map into two tiers: *réseaux d'éducation prioritaire* (REP) and *réseaux d'éducation prioritaire renforcé* (REP+), the latter reserved for schools facing the most severe concentrations of disadvantage. Classification is based on a social index combining the share of students from disadvantaged backgrounds, scholarship recipients, students residing in priority neighborhoods, and students with academic delay at entry to lower secondary school. As of 2023, priority education comprised 4,136 primary schools classified as REP and 2,458 classified as REP+, serving alongside 731 and 362 lower secondary schools respectively. Both REP and REP+ status come with additional resources for teachers, including financial bonuses, faster accumulation of points toward a school transfer, extra paid time for collaborative work, and smaller class sizes. REP+ schools receive more of these resources than REP schools, reflecting their greater level of disadvantage.

This paper focuses on two of these instruments: financial bonuses and class size reduction.

2.2 Teacher Bonus

Financial bonuses for teachers in disadvantaged schools were introduced in September 1990 as part of the original ZEP program, marking the first source of differential compensation across schools in an otherwise centrally administered, seniority-based pay system (28). These bonuses were initially small: fixed monthly amounts of approximately €25 in 1990, roughly doubled to €52 in 1991, and increased again to €79 in 1992 (28). Because they were flat amounts rather than proportional increases, the bonuses represented a larger relative wage increase for junior teachers than for senior ones. The bonus persisted through the subsequent reorganizations of the priority education map (RAR, ECLAIR), reaching approximately 1,156€ per year prior to the 2015 reform, applied uniformly across the priority education schools that would later be reclassified as REP or REP+.

The 2014–2015 reform substantially increased and differentiated the bonus by network tier. As shown in Figure 2.1, the annual bonus was set at 1,734€ for REP teachers and 2,312€ for REP+ teachers from September 2015, paid monthly on top of regular salary, and has remained unchanged for REP teachers since. The REP+ bonus was subsequently increased on three occasions: to 3,479€ in 2018, to 4,646€ in 2019, and to a fixed component of 5,114€ in 2021, the latter accompanied by an additional variable component of up to 702€ tied to school-level collective performance. The bonus is tied to the position rather than to a specific individual: any teacher exercising their function in an eligible school receives it, including replacement teachers covering for an absent colleague, who receive the bonus for the duration of the replacement while the absent teacher's payment is suspended.

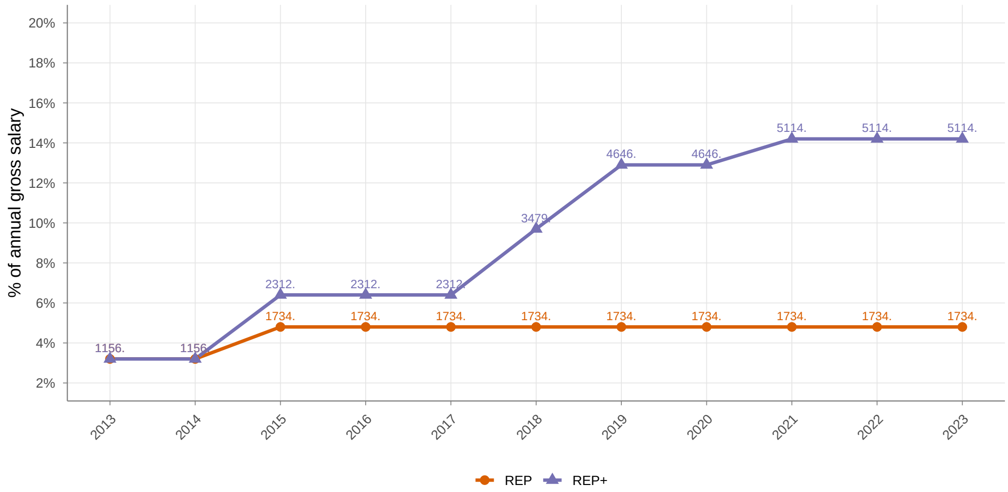


Figure 2.1: Priority education bonus as a share of gross annual salary

Note: Bonus amounts expressed as a percentage of an assumed average gross annual salary of 36,000€ for a tenured primary school teacher. REP bonus fixed at 1,734€ since 2015. REP+ bonus increased in 2018, 2019, and 2021; figure excludes the variable performance-based component introduced in 2021 (up to 702€ annually). Sources: décret n°2015-1087; Éducation.gouv.fr.

2.3 Class Size Reduction

Beginning in the 2017 school year, France implemented a large-scale class size reduction policy in priority education, capping first-grade classes in REP+ schools at approximately 12 students, roughly half the pre-reform average. The measure was extended to second grade in REP+ schools the following year, then to first grade in REP schools in the same year, and to second grade in REP schools in 2019. From 2020, the policy was further extended to the final year of pre-primary school (*grande section*) in both networks, though at a smaller magnitude and phased in more gradually than the grade 1 and grade 2 reductions. Figure 2.2 shows the resulting evolution in average class size by grade and network.

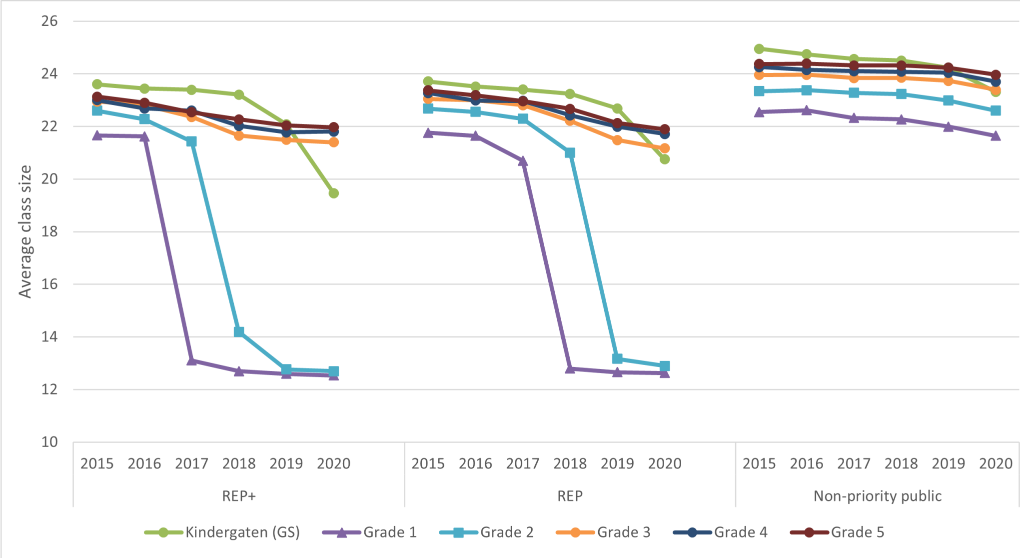


Figure 2.2: Evolution of average class size, 2015–2020

Note: Grade 1 and grade 2 class sizes fall sharply in REP+ from 2017 and in REP from 2018, converging to approximately 12–13 students, while grades 3 through 5 and non-priority classes decline only gradually. Kindergarten (grande section) reduction begins in 2020 and continues through subsequent years at a smaller magnitude.

The class size reduction did not emerge in a staffing vacuum. Since 2012, priority education schools had benefited from the *Plus de maîtres que de classes* program, which assigned supplementary teachers to support colleagues flexibly with specific subjects or small groups of students, without a formal prior evaluation of its effectiveness (9). This program was scaled up in priority education networks from 2015 before being progressively phased out from 2017, with a substantial share of its positions redeployed to staff the newly created reduced-size classrooms. The share of primary education teachers assigned to priority education schools rose from 21.7% at the start of the 2015 school year to 24.7% in 2022, an increase the Cour des Comptes attributes to the class size reduction (9).

2.4 Teacher Mobility in the French Employment System

France's public school teacher workforce operates under a career-based employment system, in which teachers are recruited through national competitive examinations and are expected to remain within the public education service for the duration of their careers, in contrast to position-based systems where movement between teaching and other occupations is common (28). For primary school teachers, the relevant examination is the *concours de recrutement de professeurs des écoles* (CRPE); successful candidates undergo a probationary training year before being granted tenure (*titularisation*).

Tenured primary school teachers are assigned to schools through a centralized, points-based system (*barème*) operated at two levels: an inter-departmental phase, through which teachers may request a transfer to a different *département*, and an intra-departmental phase, through which teachers request a specific school assignment within their department. Requests are submitted annually through the ministry's online platform (SIAM), and vacant posts are allocated to the highest-ranked candidate for each request, based on a points score. Points primarily accrue with seniority, both in the profession and in the current post, alongside smaller adjustments for family circumstances and other regulatory bonifications. Because the points scale is weighted so heavily toward seniority, newly tenured teachers typically have limited ability to secure their preferred assignment, while more senior teachers are increasingly able to move once they accumulate sufficient points.

School principals have essentially no discretion over which teachers are assigned to their school under this system. The resulting mobility pattern is well documented for secondary education and plausibly extends to the primary sector studied here: newly tenured teachers are disproportionately assigned to less sought-after posts, including those in priority education schools, while mobility becomes possible only once sufficient seniority points have accumulated (28).

A second feature of this system is central to the interpretation of the results in this paper. Non-tenured personnel (teachers in provisional posts, replacement teachers, and contract staff) absorb much of the flexibility the school system requires, and can be reassigned across schools according to staffing needs. Tenured teachers on a permanent post, by contrast, are rarely reassigned against their will. This implies that the observed mobility of tenured teachers is a reasonable proxy for their revealed preferences, even in the absence of direct data on mobility requests. This paper accordingly restricts its main analysis to tenured teachers on a permanent post (*titulaires sur poste définitif*), for whom remaining in or leaving a school reflects a deliberate choice rather than an administrative reassignment. This restriction and its implications for the interpretation of teacher mobility as a behavioral outcome are discussed further in Section 3.

3 Data

3.1 The Teacher Panel

The analysis uses the *Base Statistique des Agents* (BSA), an administrative panel recording the population of civil servants employed by the French Ministry of National Education. BSA covers all primary and secondary school teachers, along with other education personnel, and records for each individual and each year their administrative status (tenured, provisional, replacement, trainee, or contract), their school of assignment, their seniority both as a civil servant and within their current post, and basic demographic characteristics including age and sex. Because each teacher is identified by a unique national identifier across years, the panel allows any given teacher to be followed over their entire career: whether they remain in the same school, move to a different one, change administrative status, or exit the teaching profession entirely. This individual-level continuity is what makes it possible to construct the school-level retention outcome used throughout this paper. The panel used here spans 2009 to 2023.

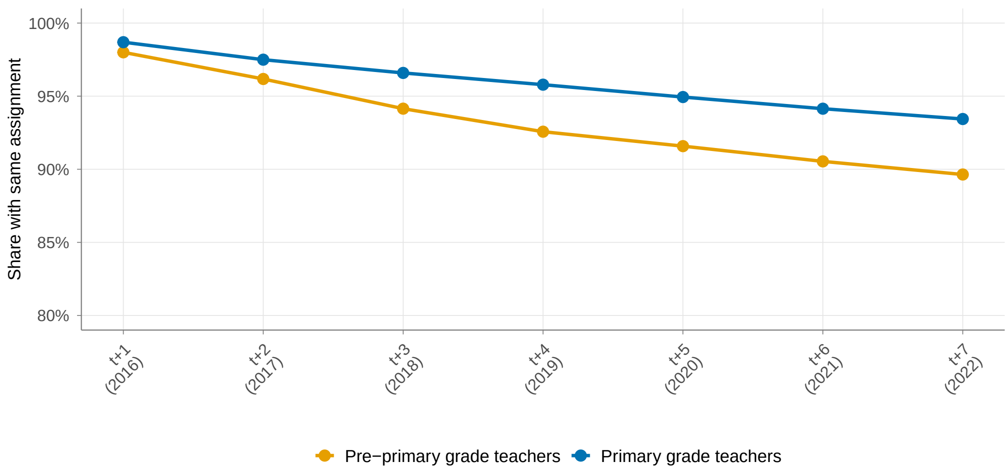
While BSA does not record the specific grade a primary school teacher is assigned to teach, it does record a broader assignment variable indicating whether a teacher's post corresponds to pre-primary teaching (ages 3–6) or primary teaching (ages 6–11), referred to throughout this paper as pre-primary and primary grade teachers respectively. This distinction forms the basis of the identification strategy described in Section 4. This assignment variable is only available in BSA from 2015 onwards, so recovering it for earlier years required an additional construction step, described below.

Constructing the assignment variable

For each teacher present in the panel in 2015 with a recorded pre-primary or primary assignment, that 2015 assignment is fixed and applied to every year of the panel, including years both before and after 2015. Teachers not present in 2015, or present but without a recorded assignment that year, are instead assigned their first directly observed pre-primary or primary assignment in a subsequent year.

This extrapolation relies on the assumption that a teacher's pre-primary or primary assignment is stable over time. I verify this by testing assignment persistence among tenured teachers on a permanent post: among teachers observed in a pre-primary or primary assignment in 2015, Figure 3.1 shows the share still in the same assignment type in each subsequent year through 2022, the final year covered by the main analysis. Persistence is high and stable: seven years after 2015, close to 90% of pre-primary grade teachers and around 94% of primary grade teachers remain in the same assignment type, a pattern consistent across REP+, REP, and non-priority schools, supporting the extrapolation of the teaching assignment variable.

Figure 3.1: Persistence of teaching assignment among tenured teachers



Note: Among tenured teachers on a permanent post with a pre-primary or primary assignment in 2015, share still in the same assignment in each following year through 2022. Pooled across REP+, REP, and non-priority schools.

Priority education classification

A parallel construction step applies to the classification of schools as REP, REP+, or non-priority. The current REP/REP+ structure was established in 2015, replacing an earlier system based on ZEP, RRS, and ECLAIR classifications, with REP largely succeeding RRS-classified schools and REP+ succeeding ECLAIR-classified schools. Of the approximately 1,000 lower secondary schools (*collèges*) that anchor priority education networks, 208 newly entered priority education status at the 2015 reform while 189 exited, with the large majority of remaining schools keeping a comparable classification (30). No national reorganization of the priority education map has taken place since 2015, making it the most recent classification available for the full period studied. I therefore record each school's REP, REP+, or non-priority status as observed in 2015 and extrapolate it to all other years, both before and after. This means that treatment status in this paper reflects each school's classification under the 2015 reform throughout the entire period studied.

Recovering teaching grade

To identify teachers assigned specifically to first- or second-grade classrooms (the grades directly targeted by the class size reduction) this paper additionally draws on DIAPRE, a separate administrative database recording student-level enrollment by grade and class, consistently available since 2017, with an earlier exception for 2010. Because DIAPRE also records each student's prior-year school information, class-level information for 2009 and 2016 could additionally be reconstructed from the 2010 and 2017 records, respectively. This data is populated by school principals and does not record each student's teacher. However, principals occasionally include the teacher's name in the class label field, and, where this occurs, teacher names can be matched to BSA records to recover the specific grade taught.

This matching is only possible where principals chose to record a teacher's name, and coverage therefore varies by year. Table 3.1 reports the resulting coverage rate by year and EP group, ranging from approximately 33% to 54% across years for which matching data are available. Coverage is similar across REP+, REP, and non-priority schools within each year, providing no indication of systematic bias toward any one group. As a result, teaching grade can be recovered for a subsample of teachers. This subsample is used in Appendix A.3 to provide descriptive evidence on the retention of first- and second-grade teachers specifically, complementing the main analysis, which relies on the broader pre-primary/primary distinction available for the full sample.

Table 3.1: Coverage of the DIAPRE grade-matching subsample by EP group, 2009–2022

Year	Non-priority	REP	REP+
2009	38.0%	33.1%	34.1%
2010	43.7%	37.5%	39.8%
2016	45.4%	41.7%	41.9%
2017	52.1%	48.1%	51.7%
2018	51.1%	50.4%	53.5%
2019	47.7%	47.8%	48.7%
2020	46.7%	46.7%	48.1%
2021	45.7%	46.9%	47.7%
2022	43.9%	45.5%	46.3%

Note: Share of tenured primary grade teachers in the analysis sample for whom teaching grade could be recovered through the DIAPRE matching procedure, by year and EP group. No matching data are available for 2011–2015. Source: DEPP, BSA-DIAPRE matched panel.

3.2 Descriptive Sample

The retention outcome used throughout this paper captures effective mobility, whether a teacher remains in or leaves a school, rather than preferred mobility. To construct a measure that most closely approximates teachers' preferences, the analysis focuses on tenured teachers holding a permanent post (*titulaires sur poste définitif*): teachers whose departure from a school is a voluntary act requiring an active mutation request, as opposed to teachers in provisional posts, replacement zones, or initial training, whose assignment is partly or fully determined administratively.

The sample is restricted to schools present in every year of the balanced panel, 2009–2023, and further restricted to schools staffed exclusively by pre-primary grade teachers or exclusively by primary grade teachers, avoiding the assignment of the same school-level retention rate to both groups. The resulting sample comprises 30,005 schools and 312,246 tenured teachers over the 2009–2023 period.

Table 3.2 reports baseline characteristics of the analysis sample in 2014, the year before the reform. Several patterns are worth noting. Priority education schools are staffed by younger and less experienced teachers than non-priority schools across both groups:

REP+ primary grade teachers are on average 39.8 years old with 13.2 years of seniority, compared to 43.9 years and 18.0 years in non-priority primary schools. The teaching workforce is highly feminized across all groups, particularly among pre-primary grade teachers, where the share of female teachers exceeds 89% in all EP groups. Within each EP group, primary grade teachers are consistently younger, less experienced, and less likely to be female than pre-primary grade teachers.

Table 3.2 also reports pre-reform annual turnover by EP group and school category in 2014, defined as one minus the school-level retention rate. In priority education schools, primary grade teachers turn over at higher rates than pre-primary grade teachers, in both REP (13.8% vs 13.2%) and REP+ schools (13.9% vs 12.9%). This differential is central to interpreting the differential responses to the reform documented in Section 5.

Table 3.2: Baseline characteristics of tenured teachers by EP group and school category, 2014

EP Group	School Category	N teachers	N schools	Share female	Mean age	Mean seniority	Turnover
Non-priority	Pre-primary	38,696	10,079	92.5%	45.4	19.8	12.4%
	Primary	85,970	14,523	80.8%	43.9	18.0	11.5%
REP	Pre-primary	8,326	1,575	90.7%	43.0	16.9	13.2%
	Primary	15,525	1,802	76.6%	40.5	14.2	13.8%
REP+	Pre-primary	5,416	983	89.8%	42.5	16.3	12.9%
	Primary	9,959	1,001	73.7%	39.8	13.2	13.9%

Note: Sample restricted to tenured teachers on permanent posts in schools staffed exclusively by pre-primary or exclusively by primary grade teachers. School category based on assignment observed in 2015. Seniority measured in years as a civil servant teacher. Turnover defined as one minus the school-level retention rate. Share female expressed as a proportion of teachers in each group. Source: DEPP, BSA panel, 2014.

4 Empirical Strategy

I estimate the causal effect of the bonus and class size reduction on teacher retention by exploiting the staggered timing and differential intensity of the two reforms: the financial bonus, introduced in 2015 and subsequently increased for REP+ schools in 2018, 2019, and 2021, and the class size reduction, progressively rolled out in first- and second-grade classrooms of REP+ schools from 2017 and REP schools one year later. The identification strategy relies on the fact that, although each teacher's exact grade is unobserved, the data record whether a teacher is a pre-primary grade teacher (ages 3–6) or a primary grade teacher (ages 6–11). Because class size reductions targeted first- and second-grade classrooms, they applied to primary grades from 2017 onwards but left pre-primary grades largely unaffected until 2020. By contrast, the bonus applied equally to teachers teaching both pre-primary and primary grades. This creates a quasi-experimental design in which pre-primary grade teachers were exposed to the bonus only, while primary grade teachers were exposed to both the bonus and class size reduction, with the intensity of both instruments differing depending on whether they taught in REP+ or REP schools. I use an event study design comparing the evolution of school-level retention rates in REP+ and REP schools with retention rates in non-priority schools, estimated separately for pre-primary grade teachers and for primary grade teachers.

4.1 Event Study Specification

The analysis compares annual retention rates of tenured teachers on a permanent post in REP+ or REP schools (treated) with retention rates in non-priority schools (never treated), separately for pre-primary and primary schools. Retention in year t is defined as the share of tenured teachers present in school s in year t who remain in the same school in year $t+1$, computed at the school level. The analysis sample is restricted to schools staffed exclusively by pre-primary grade teachers or exclusively by primary grade teachers, which avoids assigning the same school-level retention rate to both groups. Non-priority schools remained untreated throughout the period.

I estimate the following event study specification:

$$R_{st} = \alpha + \sum_{\substack{t=2009 \\ t\neq 2014}}^{2022} \tau_t \left(\text{Treated}_s \times 1[\text{Year}_t = t]\right) + \lambda_s + \delta_t + \varepsilon_{st} \tag{4.1}$$

where:

- R_{st} is the annual retention rate in school s in year t, defined as the share of tenured teachers present in year t who remain in the same school in year $t+1$
- Treated_s is an indicator equal to one for REP+ schools (or, in the corresponding specification, REP schools)

- $t = 2014$ is the omitted reference year, the last year before the 2015 bonus reform
- λ_s are school fixed effects, absorbing time-invariant differences in retention rates between treated and non-priority schools
- δ_t are year fixed effects, controlling for common time trends in retention rates affecting all schools
- ε_{st} is the error term

All specifications are weighted by the number of teachers in school s in year t. Standard errors are clustered at the school level to account for within-school correlation in retention rates over time.

The coefficients τ_t capture the treatment effect in year t, measured as the difference in retention rates between treated and non-priority schools relative to the reference year 2014, conditional on school and year fixed effects. For $t < 2015$ (pre-treatment years 2009–2013), τ_t provides a test of the parallel trends assumption: these coefficients should be statistically indistinguishable from zero if treated and non-priority schools followed similar retention trends before the reform. For $t \geq 2015$ (post-treatment years), τ_t estimates the dynamic treatment effect of the reform package, allowing the retention response to evolve over time as the policy was progressively expanded.

An important interpretive caveat applies throughout: R_{st} is a school-level aggregate, not an individual teacher's probability of remaining. The estimated coefficients therefore describe how the share of teachers retained in a school changes with treatment, not how any given teacher's individual likelihood of staying responds to the reform. This aggregate outcome reflects the combined result of individual teachers' decisions and, to the extent it exists, any residual administrative constraint on mobility within the sample studied. Section 3 discussed why this school-level measure is nonetheless informative about teacher preferences, given the specific sample of teachers analyzed.

4.2 Identification

The event study estimator identifies causal effects under the parallel trends assumption: absent the reform, retention rates in treated and non-priority schools would have followed the same evolution over time. Formally, this requires:

$$\begin{aligned} &\mathbb{E}[R_{st}(0) \mid \text{Treated}_s = 1, t] - \mathbb{E}[R_{st}(0) \mid \text{Treated}_s = 1, t-1] \\ &\quad = \mathbb{E}[R_{st}(0) \mid \text{Treated}_s = 0, t] - \mathbb{E}[R_{st}(0) \mid \text{Treated}_s = 0, t-1] \end{aligned} \tag{4.2}$$

where $R_{st}(0)$ denotes the potential retention rate in the absence of treatment.

The parallel trends assumption is plausible in this setting for two reasons. First, primary schools do not select into REP or REP+ status directly: classification is inherited from the lower secondary school (*collège*) to which each primary school is administratively attached, itself determined by a socioeconomic index computed at the network level. Primary school principals and teachers play no role in this classification, limiting the scope for strategic selection into treatment status. Second, the timing of both reforms relative to the annual teacher mobility calendar limits the scope for anticipatory behavior. The 2015 bonus was formalized by decree on 28 August 2015, effective from 1 September 2015, only days before the start of the school year and after the spring mutation campaign (through which tenured teachers request and are assigned to school posts for the following year) had already concluded. The 2017 class size reduction was announced only after the May 2017 presidential election, again after that year's mutation campaign had already taken place. In both cases, teachers' assignment decisions for the relevant school year were finalized before the reform was publicly known, leaving little scope for the observed retention patterns to reflect anticipatory mobility choices.

I assess the parallel trends assumption graphically, by plotting the event study coefficients τ_t over time and examining whether pre-treatment coefficients are close to zero and individually insignificant, and formally, through joint F-tests of pre-treatment coefficients over 2009–2013. Results of these tests are reported alongside the main results in Section 5 and summarized in Appendix Table A.1.

5 Results

The findings are presented in three parts. I first compare pre-primary grade teachers in REP+ and REP schools (treated group) to those in non-priority schools (never treated), isolating the bonus effect in the absence of any class size treatment. I then compare primary grade teachers in REP schools to those in non-priority schools, where the only policy change after 2015 was the class size reduction introduced in 2018, identifying its effect on retention. Finally, I examine primary grade teachers in REP+ schools, who received both instruments simultaneously, documenting the combined effect.

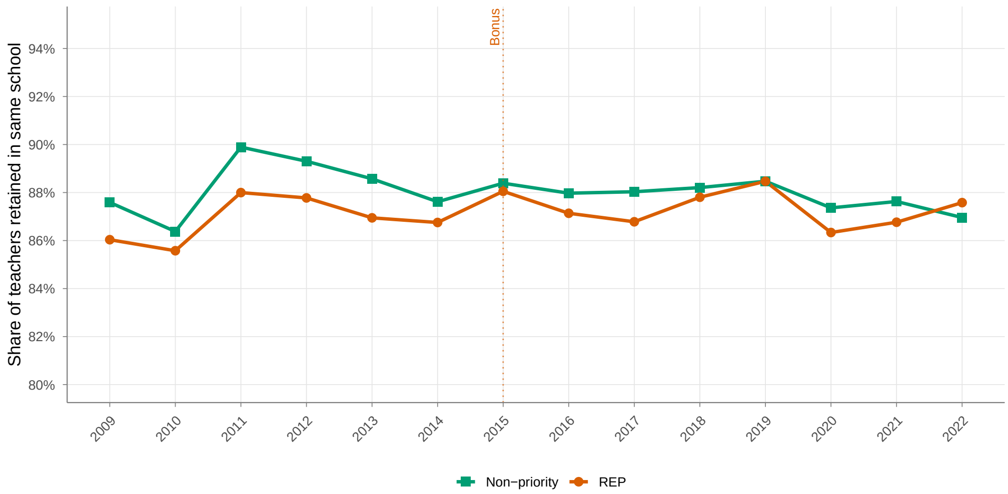
5.1 Effect of the Bonus

To isolate the bonus effect, I compare pre-primary grade teachers in REP and REP+ schools to teachers in non-priority schools. This setting also offers the opportunity to assess whether the effect depends on the magnitude of the bonus: pre-primary grade teachers in REP+ schools benefited from successive increases from an annual bonus of 2,312€ in 2015 to 5,114€ by 2021, while those in REP schools saw no further increase after 2015. I start with the REP comparison, where the bonus remained flat, before turning to REP+ where the bonus increased substantially.

Pre-primary grade teachers in REP schools

Figure 5.1 shows the evolution of retention rates for tenured pre-primary grade teachers in REP and non-priority schools. Throughout the period, retention rates in REP schools remain consistently below those of non-priority schools. A slight increase is visible in 2015 following the bonus introduction, and again in 2019, but both are followed by declines that bring REP retention back below non-priority levels. Only in 2022 does REP retention marginally exceed that of non-priority schools. The 2015 bonus reform does not appear to have generated a sustained effect on pre-primary grade teachers in REP schools.

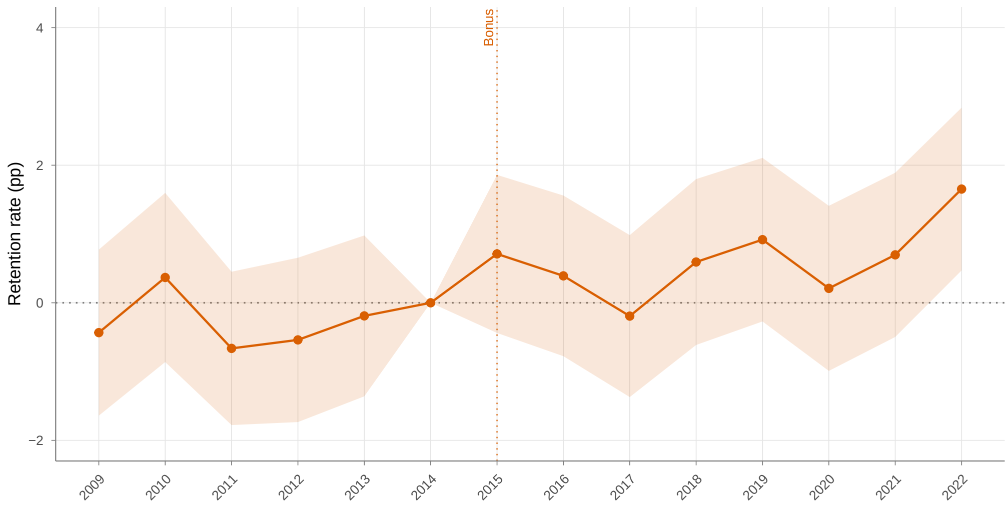
Figure 5.1: Teacher retention rates: REP vs non-priority, pre-primary grades



Note: School-level retention rate of tenured pre-primary grade teachers on permanent posts, defined as the share of teachers present in a school in year t still present in the same school in year $t+1$, averaged across REP and non-priority schools. The dotted vertical line marks the 2015 bonus reform.

These patterns are corroborated by the event study estimates presented in Figure 5.2. Pre-trend coefficients are jointly insignificant (F = 0.839, p = 0.522), and individual coefficients for 2009–2013 are small and statistically indistinguishable from zero, supporting the parallel trends assumption. No significant retention effect is detectable across the entire 2015–2021 period. A significant effect appears only in 2022 (1.7 pp, $p < 0.01$), seven years after the 2015 bonus reform. As documented in Section 2, class size reductions were progressively extended to the last grade of pre-primary education in REP and REP+ schools from 2020 onwards. The 2022 effect cannot therefore be interpreted as a causal impact of the bonus alone, as it coincides with this gradual class size reduction. Taken together, these results indicate that the REP bonus generated no detectable retention response among pre-primary grade teachers in the years following the bonus reform.

Figure 5.2: Teacher retention: REP vs non-priority, pre-primary grades



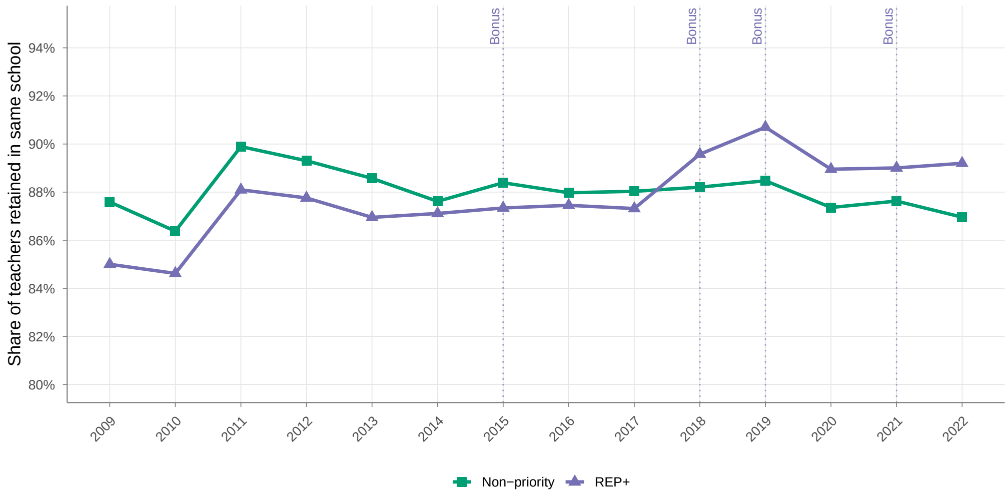
Note: Event study coefficients $\hat{\tau}_t$ from equation (4.1). All specifications include school and year fixed effects, weighted by the number of teachers, with standard errors clustered at the school level. 95% confidence intervals displayed. Reference year: 2014. Full regression tables are reported in Appendix Table A.2. The dotted vertical line marks the 2015 bonus reform.

These results establish a first benchmark: a bonus of 1,734€ annually, representing a substantial increase over the pre-2015 level but left unchanged thereafter, is insufficient to alter the retention decisions of pre-primary grade teachers. Whether a higher and increasing bonus generates a different response is examined next.

Pre-primary grade teachers in REP+ schools

Figure 5.3 shows the evolution of retention rates for tenured pre-primary grade teachers in REP+ and non-priority schools. Throughout the pre-reform period, retention rates in REP+ schools are consistently lower than in non-priority schools, though both groups follow a broadly similar pattern. A slight dip in non-priority retention in 2014 is followed by a recovery in 2015, and both groups track each other closely through 2017 with no discernible divergence following the 2015 bonus. From 2018, retention in REP+ schools not only increases sharply but exceeds that of non-priority schools for the first time, coinciding with the first bonus increase. Retention in REP+ schools rises further in 2019, following the second bonus increase, and stabilizes above the non-priority rate through 2022. The 2021 bonus increase does not generate a further rise in retention, but REP+ schools continue to retain pre-primary grade teachers at higher rates than non-priority schools through 2022. Descriptively, this pattern already suggests a retention effect coinciding with the successive bonus increases from 2018 onwards.

Figure 5.3: Teacher retention rates: REP+ vs non-priority, pre-primary grades

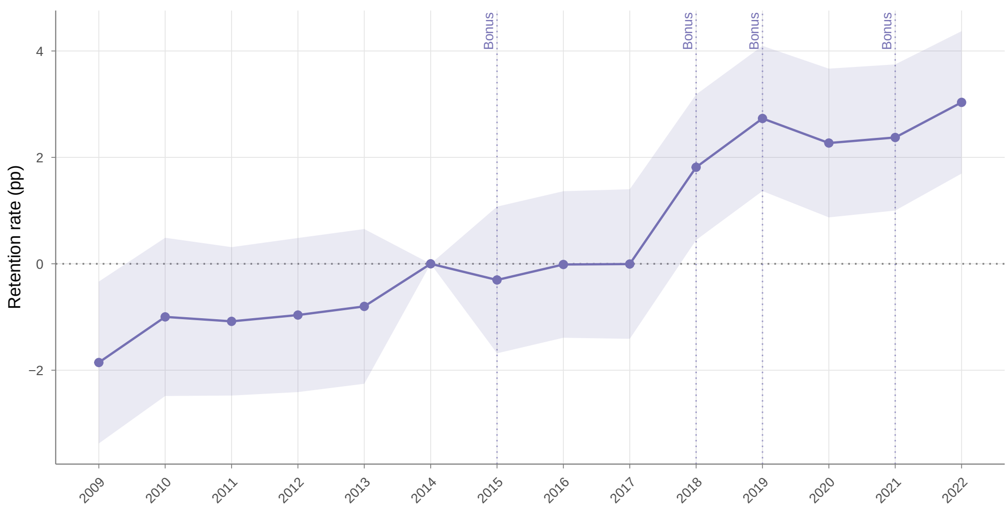


Note: School-level retention rate of tenured pre-primary teachers on permanent posts, defined as the share of teachers present in a school in year t still present in the same school in year $t+1$, averaged across REP+ and non-priority schools. Dotted vertical lines mark bonus increases for REP+ teachers.

Figure 5.4 presents the event study estimates. Pre-trend coefficients are jointly insignificant ($F = 1.186$, $p = 0.313$). Individual coefficients for 2010–2013 are small and statistically indistinguishable from zero, though a marginally significant deviation is observed in 2009 ($\hat{\tau}_{2009} = -1.86$ pp, $p < 0.05$). Given the slight divergence in non-priority retention rates in 2014 visible in Figure 5.3 and as a robustness check, I use 2013 as an alternative reference year. The 2009 coefficient becomes insignificant ($p = 0.166$) and pre-trend coefficients are jointly insignificant across all pre-reform years. Post-reform estimates are quantitatively unchanged under this alternative specification.

Turning to the post-reform period, the 2015 bonus introduction generates no detectable effect, nor do the following two years. A significant retention effect emerges in 2018, coinciding with the first bonus increase for REP+ teachers from 2,312€ to 3,479€ annually: the estimated effect is 1.81 pp ($p < 0.01$). The effect grows in 2019 following the second increase to 4,646€: 2.73 pp ($p < 0.001$). Retention gains remain significant through 2022, reaching 3.03 pp ($p < 0.001$).

Figure 5.4: Teacher retention: REP+ vs non-priority, pre-primary grades



Note: Event study coefficients $\hat{\tau}_t$ from equation (4.1). All specifications include school and year fixed effects, weighted by the number of teachers, with standard errors clustered at the school level. 95% confidence intervals displayed. Reference year: 2014. Full regression tables are reported in Appendix A. Dotted vertical lines mark bonus increases for REP+ teachers.

These results suggest that the financial bonus generates significant retention gains among pre-primary grade teachers only when it reaches a sufficient share of their compensation. Taking the average gross annual salary of a *professeur des écoles* of approximately 36,000€ (estimated from the average net monthly salary of 2,425€ documented by DEPP (11)) the REP+ bonus generated no detectable retention response at its initial 2015 level of 2,312€ (approximately 6% of gross annual salary). A significant effect emerges only from 2018, when the bonus reached 3,479€, representing approximately 10% of gross annual salary. These findings point to a threshold effect: below a certain share of compensation, the financial incentive does not alter the retention decisions of pre-primary grade teachers.

5.2 Effect of Class Size Reduction

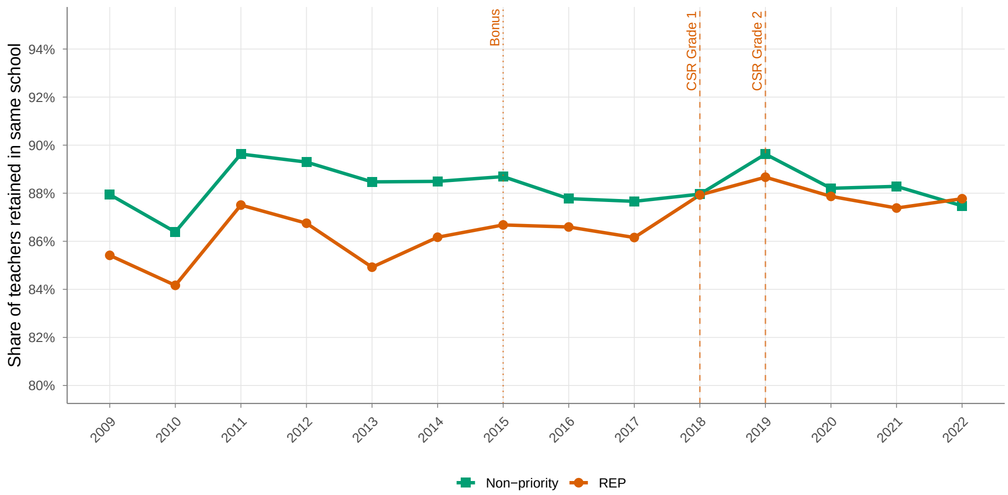
To identify the effect of class size reduction, I compare primary grade teachers in REP and REP+ schools to teachers in non-priority schools. Primary grade teachers are directly exposed to class size reductions, which halved first- and second-grade class sizes from approximately 22 to 13 students. The REP comparison offers the cleaner identification:

the REP bonus remained flat at 1,734€ throughout the period, the same level shown in the previous section to generate no detectable retention response among pre-primary grade teachers. Any retention effect observed for primary grade teachers in REP schools after 2018 could therefore be attributed to the class size reduction rather than any change in financial incentives. The REP+ comparison, where both the bonus increased substantially and class sizes were reduced, documents the combined effect of the two instruments.

Primary grade teachers in REP schools

Figure 5.5 shows the evolution of retention rates for tenured primary grade teachers in REP and non-priority schools. Throughout the pre-reform period, retention rates in REP schools remain consistently below those of non-priority schools, with both groups following a broadly similar pattern. A slight narrowing of the gap is visible around 2015 following the bonus reform, but the gap persists through 2017. From 2018, when first-grade class sizes were reduced in REP schools, retention rises and converges toward non-priority levels. A second increase is visible in 2019 following the second-grade reduction. By 2019, retention in REP schools reaches the level that non-priority schools displayed in the pre-reform period, closing the gap that had persisted throughout the pre-reform years. This convergence persists through 2022.

Figure 5.5: Teacher retention rates: REP vs non-priority, primary grades

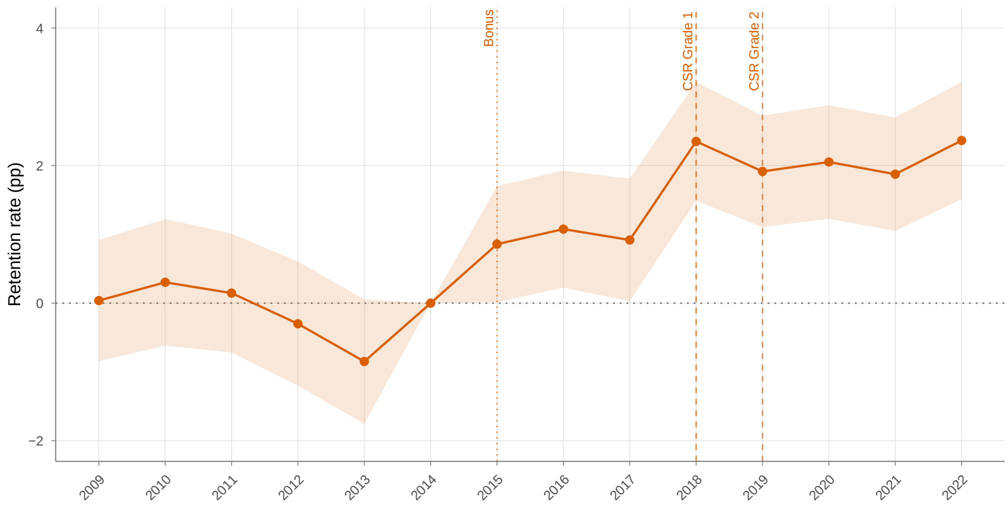


Note: School-level retention rate of tenured primary grade teachers on permanent posts, defined as the share of teachers present in a school in year t still present in the same school in year $t + 1$, averaged across REP and non-priority schools. The dotted vertical line marks the 2015 bonus reform. Dashed vertical lines mark the class size reductions in first grade (2018) and second grade (2019) for REP schools.

Figure 5.6 presents the event study estimates. Pre-trend coefficients are jointly insignificant ($F = 1.481$, $p = 0.192$), supporting the parallel trends assumption. Following the 2015 bonus reform, small positive effects emerge for primary grade teachers in REP schools relative to non-priority schools: coefficients range from 0.9 pp in 2015 to 1.1 pp in 2016 and 0.9 pp in 2017, individually significant at the 5% level. This modest initial response stands in contrast to the null effect documented for REP pre-primary grade teachers over the same period, who received the identical bonus but showed no significant retention response. This differential suggests that primary grade teachers are more responsive to a given financial incentive than pre-primary grade teachers, plausibly because they were closer to the margin of leaving. This is consistent with the higher pre-reform turnover rates documented among primary grade teachers: 13.8% annually in REP schools, compared to 13.2% for pre-primary grade teachers.

The estimated effect is larger and more precisely estimated from 2018 onwards, when first-grade class sizes were reduced in REP schools: the coefficient reaches 2.35 pp ($p < 0.001$), and remains at similar levels through 2022, ranging from 1.87 pp to 2.36 pp, all significant at the 0.1% level. The REP bonus remained unchanged at 1,734€ throughout this period, so this strengthening cannot be attributed to any new financial incentive. This pattern is consistent with class size reduction reinforcing the retention effect of the bonus, even at a bonus level that alone generated only a modest response.

Figure 5.6: Teacher retention: REP vs non-priority, primary grades

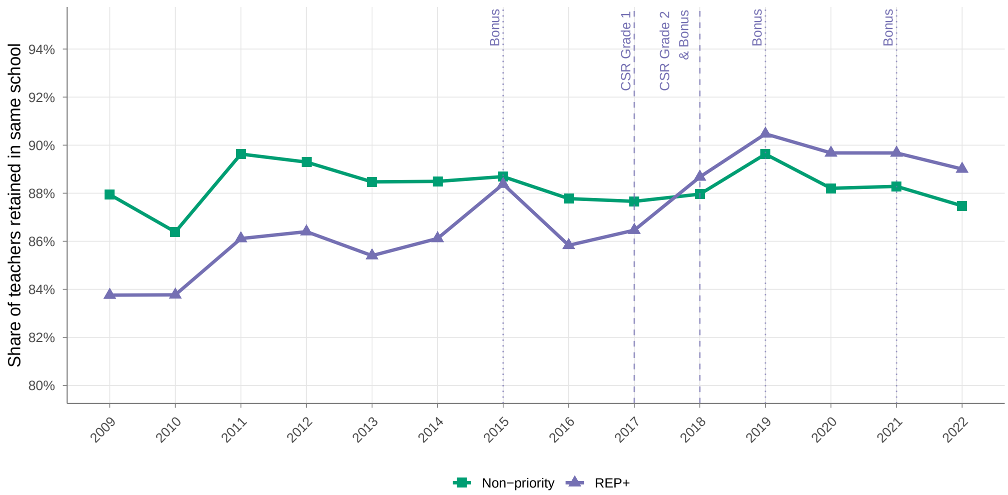


Note: Event study coefficients $\hat{\tau}_t$ from equation (4.1). All specifications include school and year fixed effects, weighted by the number of teachers, with standard errors clustered at the school level. 95% confidence intervals displayed. Reference year: 2014. Full regression tables are reported in Appendix A. The dotted vertical line marks the 2015 bonus reform. Dashed vertical lines mark the class size reductions in first grade (2018) and second grade (2019) for REP schools.

Primary grade teachers in REP+ schools

I now turn to REP+ schools, where both instruments operated simultaneously, to examine whether primary grade teachers respond more strongly when both the bonus and class size reduction are present. Figure 5.7 shows the evolution of retention rates for tenured primary grade teachers in REP+ and non-priority schools. REP+ retention remains below non-priority throughout the pre-reform period. A slight upward movement is visible from 2015, and retention rises further from 2017 when first-grade class sizes were reduced in REP+ schools. A sharp increase occurs in 2018, coinciding with both the second-grade class size reduction and the first bonus increase. REP+ retention clearly exceeds that of non-priority schools from 2018 onwards, reaching its peak in 2019–2020 and remaining elevated through 2022.

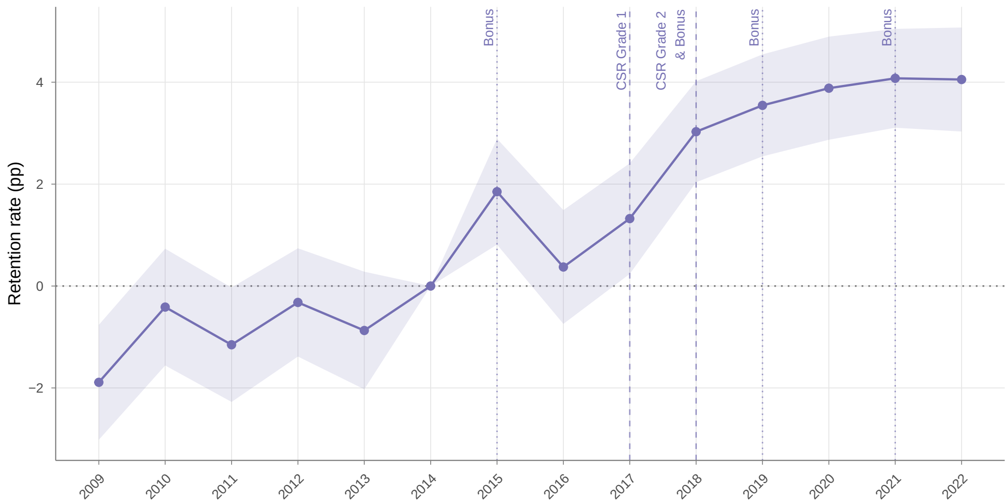
Figure 5.7: Teacher retention rates: REP+ vs non-priority, primary grades



Note: School-level retention rate of tenured primary grade teachers on permanent posts, defined as the share of teachers present in a school in year t still present in the same school in year $t + 1$, averaged across REP+ and non-priority schools. The dotted vertical line marks the 2015 bonus reform. Dashed vertical lines mark the class size reduction in first grade (2017) and the combined class size reduction in second grade and bonus increase (2018) for REP+ schools.

Figure 5.8 presents the event study estimates. Although pre-trend coefficients are jointly significant (F = 2.69, p = 0.020), this is predominantly driven by a significant deviation in 2009 ($\hat{\tau}_{2009} = -1.89$ pp, p = 0.001). Coefficients in 2012 and 2013 are small and insignificant, supporting parallel trends in the years immediately preceding treatment. Consistent with the pattern documented above, primary grade teachers in REP+ schools respond to the 2015 bonus while pre-primary grade teachers do not. Post-treatment estimates are positive and substantially larger than those for REP primary grade teachers over the same period. A significant effect emerges in 2015 (1.85 pp, $p < 0.001$), with 2016 showing a smaller and insignificant effect, before rising again in 2017 with the first-grade class size reduction (1.32 pp, $p < 0.05$) and increasing further in 2018 (3.03 pp, $p < 0.001$). Unlike the pre-primary and REP comparisons where retention gains plateau after 2019, effects continue to grow in 2021 coinciding with the third bonus increase (4.08 pp, $p < 0.001$), and remain at this level in 2022 (4.05 pp, $p < 0.001$). REP+ primary grade teachers are the only group in this analysis exposed to both a substantially increasing bonus and a class size reduction, and this comparison produces the largest retention gains observed in the paper. As in the REP comparison above, this pattern is consistent with primary grade teachers responding more strongly to the combined presence of both instruments.

Figure 5.8: Teacher retention: REP+ vs non-priority, primary grades



Note: Event study coefficients $\hat{\tau}_t$ from equation (4.1). All specifications include school and year fixed effects, weighted by the number of teachers, with standard errors clustered at the school level. 95% confidence intervals displayed. Reference year: 2014. Full regression tables are reported in Appendix A. The dotted vertical line marks the 2015 bonus reform. Dashed vertical lines mark the class size reduction in first grade (2017) and the combined class size reduction in second grade and bonus increase (2018) for REP+ schools.

5.3 Robustness to Compositional Change

A potential concern is that the observed retention gains reflect a mechanical compositional shift in the teaching workforce rather than a genuine behavioral response. This concern arises specifically in the context of class size reduction: because the policy required a substantial increase in the number of classrooms, and hence teaching positions, in treated schools, newly appointed teachers with limited seniority may have filled these positions. Under the national points-based mutation system, such teachers typically lack sufficient points to request a transfer for several years, meaning their presence in a school in a given year does not necessarily reflect a deliberate choice to stay, but rather a mechanical inability to leave. If this compositional shift is large, the retention gains estimated in the main analysis could partly or entirely reflect this mechanical effect rather than a behavioral response to the bonus or to improved working conditions.

To assess this concern, I construct a predicted retention rate for each school-year, designed to capture what retention would look like if only the seniority composition of teachers had changed, holding fixed the underlying relationship between seniority and the likelihood of staying. This is done in two steps. First, using only pre-reform years (2009–2014), I estimate the retention rate specific to each seniority level, separately for each group and grade assignment: for instance, the share of REP+ primary grade teachers with three years of seniority who remained in their school the following year, averaged over 2009–2014. This gives a fixed *seniority-retention profile* for each group and grade assignment, reflecting pre-reform retention behavior at each seniority level. Second, for every subsequent school-year, I apply this fixed profile to the actual seniority composition of teachers observed in that school and year, and average across teachers to obtain the predicted retention rate. Intuitively, this predicted rate answers the question: given who is actually teaching in this school this year, and how similarly-tenured teachers behaved before the reform, what retention rate would we expect if teachers' behavior had not changed at all after the reform?

If the observed retention gains are driven entirely by composition, actual retention should track this predicted rate closely, and controlling for the predicted rate should drive the estimated treatment effect toward zero. Table 5.1 reports estimates of the pooled post-reform effect, with and without controlling for the predicted retention rate. The coefficient changes only modestly once the predicted rate is included, indicating that the retention gains documented above reflect a genuine behavioral response rather than compositional change in the teaching workforce.

Table 5.1: Treatment effects with and without controlling for predicted retention

	Pre-primary REP+		Primary REP+		Pre-primary REP		Primary REP	
	Raw	Controlled	Raw	Controlled	Raw	Controlled	Raw	Controlled
Post-reform effect	0.0239***	0.0188***	0.0365***	0.0332***	0.0095***	0.0059*	0.0196***	0.0177***
	(0.0027)	(0.0027)	(0.0022)	(0.0022)	(0.0024)	(0.0024)	(0.0018)	(0.0018)
Predicted retention		0.9300***		0.6057***		0.8653***		0.5626***
		(0.0376)		(0.0309)		(0.0374)		(0.0306)
School FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: Dependent variable: school-level retention rate. Weighted by number of teachers, standard errors clustered at the school level in parentheses. The significant pooled effect for Pre-primary REP reflects the pooled average across all post-reform years and is driven primarily by the individually significant 2022 coefficient documented in the main event study estimates; other post-reform years show no significant effect individually. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.

Taken together, and net of any compositional change, these results establish that the bonus, when sufficiently large, increased teacher retention in priority education schools. The bonus alone generates retention gains of approximately 2–3 pp among pre-primary grade teachers in REP+ schools, where class size reduction did not apply over most of the period studied. Primary grade teachers, by contrast, show a more complex pattern: retention gains appear already at the 2015 bonus reform, before any class size change, but grow larger and more precisely estimated from 2018 onwards in both REP and REP+ schools, reaching approximately 2 pp in REP and up to 4 pp in REP+, the largest effect observed in this paper. This strengthening coincides with the introduction of class size reduction in both networks, suggesting that class size reduction contributed an additional retention effect on top of the bonus, with the largest gains observed where both instruments operated together.

6 Conclusion

These findings speak to the broader question of how education systems can retain teachers in disadvantaged schools. Higher pay can be effective, but only above a certain threshold: a modest bonus was insufficient to alter retention decisions, while a substantially larger and increasing bonus produced strong and lasting gains. Improving teachers' working environment also appears to matter: retention gains among primary grade teachers strengthen once class size reduction is introduced, suggesting that better working conditions can themselves support retention. The largest gains are observed where both instruments operate together, consistent with higher pay and better working conditions reinforcing one another. This pattern suggests that modest increases in compensation alone may do little to address teacher shortages in disadvnataged schools, and that policymakers seeking to stabilize teaching staff in these schools should consider higher pay and better working conditions together as complementary tools.

Two considerations are worth noting when interpreting the magnitude of these results. First, the class size reduction estimates are likely conservative. They are computed on the full population of tenured primary grade teachers in treated schools, while the policy applied specifically to first- and second-grade classrooms (only two out of five grade levels in a typical primary school) so the true effect on directly treated teachers is plausibly larger than what these school-level averages capture. Evidence from a sample of primary teachers for whom their teaching grade could be recovered, reported in Appendix A.3, supports this: first- and second-grade teachers, who were comparable to other primary grade teachers on observable characteristics and retention before the reform, show significantly higher survival in their school in the years following the reform, reaching 4.5 to 5.8 percentage points by 2022, while no comparable divergence emerges among primary grade teachers in non-priority schools over the same period.

Second, both the bonus and class size effects are estimated on tenured teachers holding a permanent post, the most stable segment of the primary teaching workforce, whose pre-reform retention rates already exceeded 85% annually. That measurable retention gains emerge even within this already-stable population suggests that effects on more mobile teachers may be substantially larger, though identifying such effects faces the challenge that their mobility is partly administratively determined rather than voluntary.

Future research could exploit administrative data on teacher mobility requests (*voeux de mobilité*), which record whether and where teachers request a transfer through the national mutation system, directly revealing their preferences. If such data were to become available at scale, they would allow direct estimation of the effect of higher pay and better working conditions on the revealed preferences of all teachers, including new teachers and those in provisional posts and replacement zones. This would provide a more complete picture of how education policies shape teacher retention and mobility across disadvantaged schools.

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Appendix

A Additional Results and Robustness Checks

A.1 Pre-Trend Tests

Table A.1 reports joint F-tests of pre-treatment coefficients for all main event study specifications, testing the null hypothesis that treatment and control groups followed parallel trends before each reform.

Table A.1: Joint pre-trend F-tests, main event study specifications

Comparison	Group	F-statistic	p-value
REP+ vs non-priority	Pre-primary teachers	1.186	0.313
	Primary teachers	2.690	0.020
REP vs non-priority	Pre-primary teachers	0.839	0.522
	Primary teachers	1.481	0.192
REP+ vs non-priority (ref: 2013)	Pre-primary teachers	0.557	0.694
	Primary teachers	2.190	0.067

Note: Joint F-test of the null hypothesis that all pre-treatment interaction coefficients are jointly equal to zero. Unless otherwise noted, specifications use 2014 as the reference year and test coefficients for 2009–2013. The final panel reports an alternative specification using 2013 as the reference year, testing coefficients for 2009–2012 only. All specifications include school and year fixed effects, weighted by number of teachers, with standard errors clustered at the school level.

A.2 Main Event Study Regression Tables

Tables A.2 and A.3 report full regression results for the four main event study comparisons discussed in Section 5.

Table A.2: Event study estimates, pre-primary teachers

Dependent Variable:	retention_rate	
	REP+/Non-priority	REP/Non-priority
treatment × 2009	-0.019**	-0.004
	(0.008)	(0.006)
treatment × 2010	-0.010	0.004
	(0.008)	(0.006)
treatment × 2011	-0.011	-0.007
	(0.007)	(0.006)
treatment × 2012	-0.010	-0.005
	(0.007)	(0.006)
treatment × 2013	-0.008	-0.002
	(0.007)	(0.006)
treatment × 2015	-0.003	0.007
	(0.007)	(0.006)
treatment × 2016	-0.0001	0.004
	(0.007)	(0.006)
treatment × 2017	-4.17×10^{-5}	-0.002
	(0.007)	(0.006)
treatment × 2018	0.018***	0.006
	(0.007)	(0.006)
treatment × 2019	0.027***	0.009
	(0.007)	(0.006)
treatment × 2020	0.023***	0.002
	(0.007)	(0.006)
treatment × 2021	0.024***	0.007
	(0.007)	(0.006)
treatment × 2022	0.030***	0.017***
	(0.007)	(0.006)
School FE	Yes	Yes
Year FE	Yes	Yes
Observations	154,791	163,083
R^2	0.11376	0.11464
Within R^2	0.00112	0.00023

Note: Event study coefficients from equation (4.1), weighted by number of teachers, clustered standard errors at the school level in parentheses. Reference year: 2014. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.

Table A.3: Event study estimates, primary teachers

Dependent Variable:	retention_rate	
	REP+/Non-priority	REP/Non-priority
treatment × 2009	-0.019***	0.0004
	(0.006)	(0.004)
treatment × 2010	-0.004	0.003
	(0.006)	(0.005)
treatment × 2011	-0.011**	0.001
	(0.006)	(0.004)
treatment × 2012	-0.003	-0.003
	(0.005)	(0.005)
treatment × 2013	-0.009	-0.008*
	(0.006)	(0.005)
treatment × 2015	0.018***	0.009**
	(0.005)	(0.004)
treatment × 2016	0.004	0.011**
	(0.006)	(0.004)
treatment × 2017	0.013**	0.009**
	(0.006)	(0.005)
treatment × 2018	0.030***	0.024***
	(0.005)	(0.004)
treatment × 2019	0.035***	0.019***
	(0.005)	(0.004)
treatment × 2020	0.039***	0.021***
	(0.005)	(0.004)
treatment × 2021	0.041***	0.019***
	(0.005)	(0.004)
treatment × 2022	0.041***	0.024***
	(0.005)	(0.004)
School FE	Yes	Yes
Year FE	Yes	Yes
Observations	217,335	228,567
R^2	0.14555	0.14365
Within R^2	0.00265	0.00089

Note: Event study coefficients from equation (4.1), weighted by number of teachers, clustered standard errors at the school level in parentheses. Reference year: 2014. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.

A.3 Class Size Effects on First- and Second-Grade Teachers

Section 6 notes that the main class size reduction estimates are likely conservative, as they are computed on the full population of tenured primary grade teachers while the policy specifically targeted first- and second-grade classrooms. This section reports descriptive evidence from a subsample of teachers for whom teaching grade could be recovered through the DIAPRE matching procedure described in Section 3.1. Because this procedure covers only a subset of teachers and does not allow for a formal pre-trend test comparable to the main analysis, the evidence reported here is intended as suggestive support for the dilution argument rather than as an independent causal estimate.

Baseline comparability

Table A.4 compares first- and second-grade teachers to other primary grade teachers on observable characteristics and retention in 2010, the cleanest pre-reform year for which grade-level data are available, prior to the introduction of both the bonus and class size reduction. The two groups are broadly comparable across EP groups on age, seniority, and retention, with differences of one percentage point or less in retention across all three networks. This comparability supports the interpretation of any subsequent divergence between the two groups as related to the reform rather than reflecting pre-existing differences.

Table A.4: Baseline characteristics by grade category, 2010

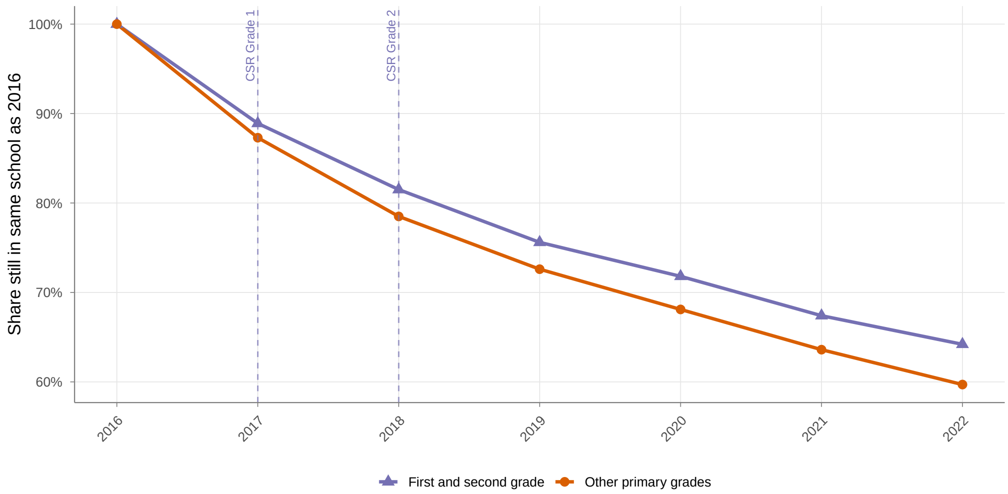
EP Group	Grade	N teachers	Share female	Mean age	Mean seniority	Retention
Non-priority	Grade 1–2	17,288	87.0%	40.7	15.2	91.2%
	Other primary	20,641	74.0%	41.5	15.7	91.0%
REP	Grade 1–2	2,716	85.9%	37.4	11.8	87.7%
	Other primary	3,168	71.3%	37.9	12.0	87.5%
REP+	Grade 1–2	1,787	82.7%	36.4	10.5	86.0%
	Other primary	2,157	70.1%	37.0	11.0	86.4%

Note: Sample restricted to tenured primary grade teachers for whom teaching grade could be recovered through the DIAPRE matching procedure. Grade 1–2 refers to teachers directly targeted by the class size reduction. Retention defined as the share of teachers present in a school in 2010 still present in the same school in 2011. Source: DEPP, BSA-DIAPRE matched panel, 2010.

Survival evidence

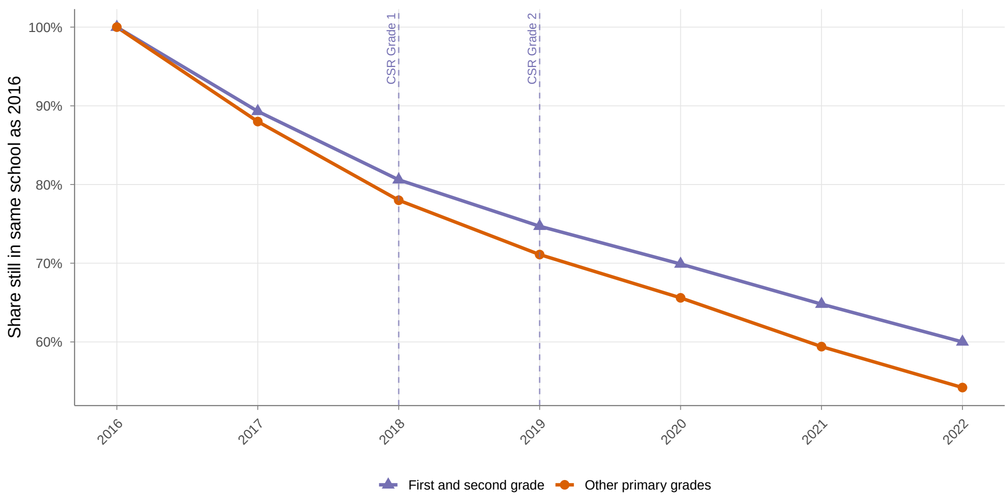
Figure A.1 and Figure A.2 show, for a cohort of teachers anchored to their 2016 teaching grade, the share still present in the same school in each subsequent year through 2022, separately for first- and second-grade teachers and other primary grade teachers, in REP+ and REP schools respectively. In both networks, first- and second-grade teachers show higher survival than other primary grade teachers from the year class size reduction was introduced, with the gap widening over subsequent years, reaching 4.5 percentage points in REP+ and 5.8 percentage points in REP by 2022. Figure A.3 reports the same comparison for non-priority schools, where a divergence of 2.4 percentage points emerges by 2022, markedly smaller than the gap observed in REP and REP+ schools.

Figure A.1: Survival of first- and second-grade teachers relative to other primary grade teachers, REP+ schools



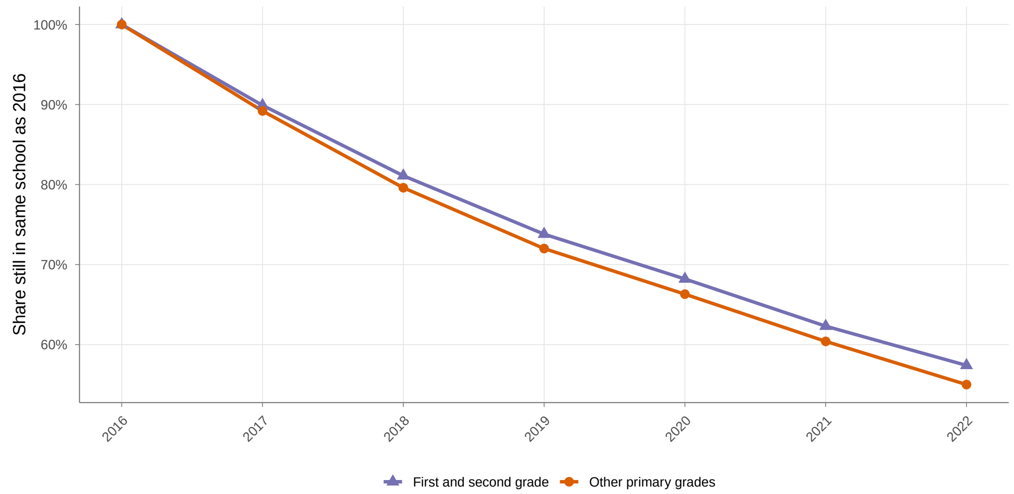
Note: Share of tenured teachers, anchored to their 2016 teaching grade, still present in their 2016 school in each subsequent year. Dashed vertical lines mark the class size reduction in first grade (2017) and second grade (2018) in REP+ schools.

Figure A.2: Survival of first- and second-grade teachers relative to other primary grade teachers, REP schools



Note: Share of tenured teachers, anchored to their 2016 teaching grade, still present in their 2016 school in each subsequent year. Dashed vertical lines mark the class size reduction in first grade (2018) and second grade (2019) in REP schools.

Figure A.3: Survival of first- and second-grade teachers relative to other primary grade teachers, non-priority schools

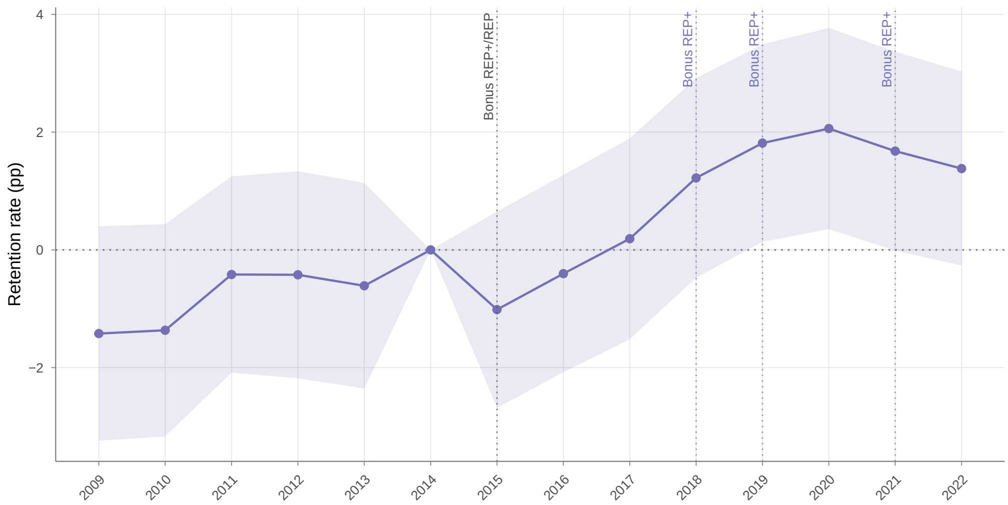


Note: Share of tenured teachers, anchored to their 2016 teaching grade, still present in their 2016 school in each subsequent year. No class size reduction applied to non-priority schools over this period.

A.4 REP+ versus REP

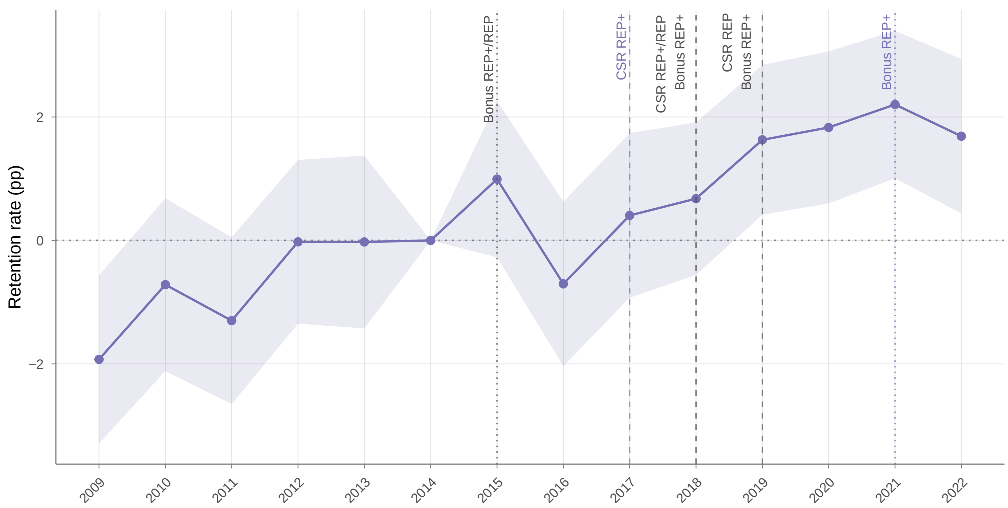
As a complementary comparison to the main results, Figures A.4–A.5 and Table A.5 report event study estimates comparing REP+ to REP schools directly. This comparison combines both the bonus differential between the two networks and the one-year timing offset in class size reduction, and is therefore harder to interpret cleanly than the comparisons to non-priority schools presented in the main text.

Figure A.4: Teacher retention: REP+ vs REP, pre-primary grades



Note: Event study coefficients comparing pre-primary grade teachers in REP+ schools to those in REP schools. All specifications include school and year fixed effects, weighted by the number of teachers, with standard errors clustered at the school level. 95% confidence intervals displayed. Reference year: 2014.

Figure A.5: Teacher retention: REP+ vs REP, primary grades



Note: Event study comparing primary grade teachers in REP+ schools to those in REP schools. All specifications include school and year fixed effects, weighted by the number of teachers, with standard errors clustered at the school level. 95% confidence intervals displayed. Reference year: 2014.

Table A.5: Event study estimates, REP+ vs REP

Dependent Variable:	retention_rate	
	Pre-primary	Primary
treatment × 2009	-0.014	-0.019***
	(0.009)	(0.007)
treatment × 2010	-0.014	-0.007
	(0.009)	(0.007)
treatment × 2011	-0.004	-0.013*
	(0.009)	(0.007)
treatment × 2012	-0.004	-0.0002
	(0.009)	(0.007)
treatment × 2013	-0.006	-0.0002
	(0.009)	(0.007)
treatment × 2015	-0.010	0.010
	(0.009)	(0.006)
treatment × 2016	-0.004	-0.007
	(0.009)	(0.007)
treatment × 2017	0.002	0.004
	(0.009)	(0.007)
treatment × 2018	0.012	0.007
	(0.009)	(0.006)
treatment × 2019	0.018**	0.016***
	(0.009)	(0.006)
treatment × 2020	0.021**	0.018***
	(0.009)	(0.006)
treatment × 2021	0.017*	0.022***
	(0.009)	(0.006)
treatment × 2022	0.014	0.017***
	(0.008)	(0.006)
School FE	Yes	Yes
Year FE	Yes	Yes
Observations	35,794	39,256
R^2	0.14673	0.20360
Within R^2	0.00165	0.00283

Note: Event study coefficients from equation (4.1), comparing REP+ schools to REP schools. Weighted by number of teachers, clustered standard errors at the school level in parentheses. Reference year: 2014. Signif. codes: ***: 0.01, **: 0.05, *: 0.1.